

Music Style Transfer and Genre Remastering

General Field Overview

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Question how do we change style while keeping musical identity recognizable?

Theme the field moved from raw generation toward controllable representations

Goal explain the concepts first, then the model families that matter most

Speaker split

Kelsey: introduction and concepts

Sarah: early generation, VAEs, MoVE, RAVE

Ahmed: AudioLM, diffusion, and field trends

Talk Map

1. Problem framing

What style transfer means in music, and why it is harder than the image analogy suggests.

2. Early field history

WaveNet, GANs, WaveGAN, GANSynth, and the limits of early audio generation.

3. Modern direction

latent audio, RAVE, AudioLM, diffusion, and why control now matters more than raw novelty

Organizing question

Every section should answer: *how does this help preserve content while changing style?*

Part I

Framing the Problem

Music style transfer field overview

What Is Style Transfer?

Image analogy

Content is the object layout and scene structure.

Style is the color palette, texture, and brush-stroke pattern.

Music version

Instead of changing color, we want to change timbre, instrumentation, texture, articulation, and production character.

source

style reference

preserve underlying structure

change the expressive surface

transfer output

Style Versus Content

Content

- melody
- rhythm
- notes being played
- broad harmonic path
- phrase contour

Style

- timbre and tone quality
- instrumentation
- texture and density
- dynamics and articulation
- expressive details

$$x_{\text{src}} \rightarrow \hat{x}_{\text{tgt}} \quad \text{with} \quad \text{content}(\hat{x}_{\text{tgt}}) \approx \text{content}(x_{\text{src}})$$

Why Music Style Transfer Is Harder Than It First Seems

Music is multilevel

Melody, rhythm, harmony, texture, instrumentation, room acoustics, mixing, and long-range form all live in the same signal.

Entanglement problem

Changing one layer often damages another. A timbre edit can distort melody, phrase identity, or realism.

The coat-of-paint failure mode

Weak systems sound processed rather than genuinely re-authored. The arrangement still feels like the source.

Genre makes this harder

Genre is not one surface sound. It is a broader manifold shaped by groove, conventions, production norms, and listener expectations.

Musical Representations

Raw waveform

highest fidelity, but a very dense space for semantic editing



Spectrograms / log-mel

expose harmonic and rhythmic structure more clearly



Symbolic formats like MIDI

easy to manipulate, but weak for final timbral realism



Learned latent spaces

modern attempt to balance control, abstraction, and fidelity

Recurring field lesson

Representation choice often matters as much as architecture choice.

Practical implication

Many important model advances are really advances in *where* generation happens.

STFT and MIDI: Two Useful Anchors

Fourier Transform and STFT

The Fourier Transform tells us which frequencies are present overall. The STFT applies it to short overlapping windows, giving a time-frequency picture of the sound.

$$X(m, \omega) = \sum_n x[n] w[n - m] e^{-j\omega n}$$

MIDI versus audio

MIDI stores musical events such as note onsets, durations, and velocities. It is compact and editable, but it does not directly capture final timbre or production character.

Why this matters

Symbolic generation is easier, but audio style transfer needs waveform realism and sonic detail.

Core Challenges in Music Style Transfer

Representation problem

what is the right working space for controllable editing?

Realism problem

even a structurally correct output fails if it sounds phasey, noisy, or synthetic

Disentanglement problem

how can content be preserved while style is changed?

Long-form coherence problem

local quality is not enough if longer outputs drift or break at chunk boundaries

Field structure

Early models struggled with representation and realism. Later work focused on disentanglement, latent controllability, and re-authoring.

Part II

Early Audio Generation and Latent Audio

Music style transfer field overview

WaveNet

What it was

One of the first major neural systems to generate realistic audio with autoregressive sample-by-sample prediction.

Core mechanism

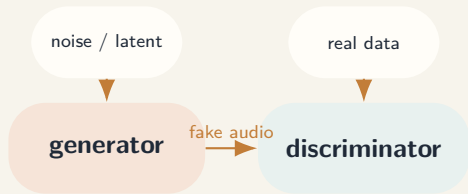
dilated causal convolutions let the model see larger temporal context without looking into the future



Why it is only background

It showed that high-quality neural audio was possible, but it was slow and not a practical direct answer to style transfer.

Background About GANs



Generator

produces candidate samples from noise or latent codes

Discriminator

learns to distinguish real from fake and pushes the generator toward realism

Historical role

GANs often produced sharper outputs than plain autoencoders, but stability remained difficult.

WaveGAN

Key idea

Adapt image GAN ideas to one-dimensional audio by using 1D convolutions, larger stride, and no batch normalization.

Phase shuffle

random phase shifts stop the discriminator from relying too heavily on trivial phase artifacts

What it showed

parallel raw-audio generation was possible

Two major limits

- phase precession made outputs sound phasey or unnatural
- fixed short clips made long-form coherent generation difficult

GANSynth

generate magnitude + IF

log-magnitude spectrograms and instantaneous frequency spectrograms

integrate IF into phase

recover unwrapped phase from a smoother representation

inverse STFT to waveform

reconstruct audio from a more structured frequency-domain description

Why it helped

Instantaneous frequency is more stable than raw phase, so the model avoids some of WaveGAN's phase problems.

Field lesson

Representation choice matters. A better working space can improve realism more than brute-force raw waveform generation.

Why Early Audio Generation Was Not Enough

WaveNet, WaveGAN, GANSynth

all helped prove that neural audio generation was possible

But they stayed weak for transfer

they could generate or condition audio, but did not explicitly guarantee content preservation

Three combined takeaways

1. raw audio generation was possible
2. representation choice matters
3. preserving content while changing style remained unsolved

VAEs and Latent Audio

$$x \xrightarrow{\text{encoder}} z \xrightarrow{\text{decoder}} \hat{x}$$

Why VAEs were attractive

A latent space gives a structured place where musical properties might be manipulated.

Why naive VAEs struggled on raw audio

- posterior collapse
- one weak global bottleneck
- sequential reconstruction difficulty

Still important

The latent-space idea remained valuable even when the first audio VAEs were weak.

MoVE

What MoVE is

Modulated Variational auto-Encoders for many-to-many musical timbre transfer

Why it matters

It was built specifically for musical timbre transfer, not just generic audio generation.

Main ingredients

- VAE latent structure
- FiLM conditioning
- many-to-many transfer in one multi-domain model
- MMD objective to align distributions

Limitation

mainly timbre transfer rather than full genre remastering

RAVE

spectral losses

compare frequency structure rather than only raw waveform samples

latent sequence over time

preserve temporal structure instead of compressing whole clips into one weak vector

adversarial decoder sharpening

improve realism after learning a compact latent representation

Why RAVE matters for style transfer

It made high-quality latent audio practical, which made editing and transfer much more realistic.

Position in the story

RAVE is a major step from generic latent modeling toward controllable neural audio.

RAVE: Caillon and Esling (2021).

Part III

Modern Controllable Synthesis

Music style transfer field overview

AudioLM and Long-Range Structure

What AudioLM does

Treat audio generation like language modeling over discrete tokens.

Important idea

Separate token streams for long-range structure and fine acoustic detail.

Why it matters here

It is not a style-transfer model directly, but it addresses one of the hardest downstream problems: long-form coherence.

Field role

evidence that modern audio models can preserve meaningful structure over longer spans

BigVGAN and FiLM

BigVGAN

A universal neural vocoder that turns intermediate representations such as mel-spectrograms into more realistic waveform audio.

$$f(x) = x + \frac{1}{\alpha} \sin^2(\alpha x)$$

Role

realism stage rather than transfer logic

FiLM

Feature-wise Linear Modulation lets a condition vector scale and shift hidden features throughout a network.

$$\text{FiLM}(F; c) = \gamma(c) \odot F + \beta(c)$$

Role

directly relevant to style-control logic

DDPM

$$q(x_t \mid x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I)$$

Core idea

Gradually destroy a sample with noise, then learn the reverse process that reconstructs it.

Why diffusion matters

- more stable than adversarial training in many settings
- iterative generation gives more room for control
- better suited to re-authoring than one-shot synthesis

Classifier-Free Guidance and SDEdit

$$\hat{\epsilon}_{\text{cfg}} = \epsilon_{\theta}(x_t, t, \emptyset) + w [\epsilon_{\theta}(x_t, t, c) - \epsilon_{\theta}(x_t, t, \emptyset)]$$

CFG

guidance scale w is a practical knob for how aggressively target style should dominate

$$x(t) = \alpha(t)x(0) + \sigma(t)z, \quad z \sim \mathcal{N}(0, I)$$

SDEdit

start from a noised version of an existing source instead of pure noise

Why these matter for style transfer

Together they make the content-versus-style tradeoff explicit rather than implicit.

AudioLDM and Why Diffusion Is the Modern Direction

AudioLDM

Latent diffusion for audio shows that high-quality controllable generation can happen in learned latent spaces rather than directly in waveform space.

Why it is relevant

It makes diffusion feel much more applicable to editing and transformation, not only text-to-audio generation.

Why diffusion leads the current story

- progressive generation
- strong conditioning
- explicit guidance control
- source anchoring through editing-style formulations

Main claim

Diffusion is central not because it is trendy, but because it matches the style-transfer problem better than many earlier families.

Overview of Field Trends

1. Disentanglement

separate what should remain stable from what should be editable



2. Controllable latent audio

move generation into spaces that are easier to manipulate meaningfully



3. Diffusion-based re-authoring

use iterative editing and guidance rather than one opaque transform



4. Long-form coherence

move beyond short local realism toward stable musical structure

The larger shift

The field moved from raw generation tricks toward structured representations, explicit control, and longer-horizon musical stability.

This is why the emphasis changed

MoVE, RAVE, AudioLM, and AudioLDM now say more about the field than a long walkthrough of older waveform internals.

Conclusion

Main lesson

Music style transfer is not just an audio-generation problem. It is simultaneously a structure-preservation, representation, conditioning, synthesis, and coherence problem.

Historical arc

Early waveform models proved that neural audio was possible. Later work made the field more controllable, more structured, and more relevant to real transfer.

Final framing

The field is moving from surface transfer toward structure-aware genre reconstruction.

Key models to remember

WaveNet, WaveGAN, GANSynth, MoVE, RAVE, AudioLM, BigVGAN, DDPM, CFG, SDEdit, AudioLDM